Functions: Composite & Inverse

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Domain, Range & Function Concepts**

The function $f(x) = \frac{1}{x+6}$ is undefined at one value of x. Find this value.

Final Answer: x=-6. The denominator $x + 6$ is zero when $x = -6$, so f is undefined there.

2 **[Foundational]** • **Target: Composite Functions**

The function $f(x) = \frac{1}{x+5}$ is undefined at one value of x. Find this value.

Final Answer: x=-5. The denominator $x + 5$ is zero when $x = -5$, so f is undefined there.

3 **[Foundational]** • **Target: Inverse Functions & Self-Inverses**

The function $f(x) = \frac{1}{x+4}$ is undefined at one value of x. Find this value.

Final Answer: x=-4. The denominator $x + 4$ is zero when $x = -4$, so f is undefined there.

4 **[Foundational]** • **Target: Graphical Relationships & Exam Challenges**

The function $f(x) = \frac{1}{x+3}$ is undefined at one value of x. Find this value.

Final Answer: x=-3. The denominator $x + 3$ is zero when $x = -3$, so f is undefined there.

5 **[Foundational]** • **Target: Domain, Range & Function Concepts**

The function $f(x) = \frac{1}{x+2}$ is undefined at one value of x. Find this value.

Final Answer: x=-2. The denominator $x + 2$ is zero when $x = -2$, so f is undefined there.

6 **[Foundational]** • **Target: Composite Functions**

The function $f(x) = \frac{1}{x+7}$ is undefined at one value of x. Find this value.

Final Answer: x=-7. The denominator $x + 7$ is zero when $x = -7$, so f is undefined there.

7 **[Foundational]** • **Target: Inverse Functions & Self-Inverses**

The function $f(x) = \frac{1}{x+9}$ is undefined at one value of x. Find this value.

Final Answer: x=-9. The denominator $x + 9$ is zero when $x = -9$, so f is undefined there.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Graphical Relationships & Exam Challenges

The function $f(x) = \frac{1}{x+8}$ is undefined at one value of x. Find this value.

Final Answer: x=-8. The denominator $x + 8$ is zero when $x = -8$, so f is undefined there.

9 [Intermediate] • Target: Domain, Range & Function Concepts

The function $f(x) = \frac{1}{x-4}$ is undefined at one value of x. Find this value.

Final Answer: x=4. The denominator $x - 4$ is zero when $x = 4$, so f is undefined there.

10 [Intermediate] • Target: Composite Functions

The function $f(x) = \frac{1}{x-8}$ is undefined at one value of x. Find this value.

Final Answer: x=8. The denominator $x - 8$ is zero when $x = 8$, so f is undefined there.

11 [Intermediate] • Target: Inverse Functions & Self-Inverses

The function $f(x) = \frac{1}{x-1}$ is undefined at one value of x. Find this value.

Final Answer: x=1. The denominator $x - 1$ is zero when $x = 1$, so f is undefined there.

12 [Intermediate] • Target: Graphical Relationships & Exam Challenges

The function $f(x) = \frac{1}{x-2}$ is undefined at one value of x. Find this value.

Final Answer: x=2. The denominator $x - 2$ is zero when $x = 2$, so f is undefined there.

13 [Intermediate] • Target: Domain, Range & Function Concepts

The function $f(x) = \frac{1}{x-3}$ is undefined at one value of x. Find this value.

Final Answer: x=3. The denominator $x - 3$ is zero when $x = 3$, so f is undefined there.

14 [Intermediate] • Target: Composite Functions

The function $f(x) = \frac{1}{x-7}$ is undefined at one value of x. Find this value.

Final Answer: x=7. The denominator $x - 7$ is zero when $x = 7$, so f is undefined there.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Inverse Functions & Self-Inverses • Cross-Lesson: 4.01 Coordinate Geometry

The function $f(x) = \frac{1}{x-5}$ is undefined at one value of x. Find this value.

Final Answer: x=5. The denominator $x - 5$ is zero when $x = 5$, so f is undefined there.

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16 [Challenging] • Target: Graphical Relationships & Exam Challenges • Cross-Lesson: 5.01 Trigonometry

The function $f(x) = \frac{1}{x-6}$ is undefined at one value of x. Find this value.

Final Answer: x=6. The denominator $x - 6$ is zero when $x = 6$, so f is undefined there.

17 [Challenging] • Target: Domain, Range & Function Concepts • Cross-Lesson: 6.01 Differentiation & Integration

Given $f(x) = 5x - 1$ and $g(x) = 2x - 5$, find $gf(x)$.

Final Answer: 10x-7. Applying Functions: Composite & Inverse principles,

$gf(x) = g(5x - 1) = 2(5x - 1) - 5 = 10x - 7$. Therefore, the verified correct option is 10x-7.

18 [Challenging] • Target: Composite Functions • Cross-Lesson: 7.01 Statistics & Probability

Given $f(x) = 2x - 5$ and $g(x) = 5x - 2$, find $fg(x)$.

Final Answer: 10x-9. Applying Functions: Composite & Inverse principles,

$fg(x) = f(5x - 2) = 2(5x - 2) - 5 = 10x - 9$. Therefore, the verified correct option is 10x-9.

19 [Challenging] • Target: Inverse Functions & Self-Inverses • Cross-Lesson: 9.01 Circular Measure

Given $f(x) = 2x + 2$ and $g(x) = 2x + 6$, find $fg(x)$.

Final Answer: 4x+14. Applying Functions: Composite & Inverse principles,

$fg(x) = f(2x + 6) = 2(2x + 6) + 2 = 4x + 14$. Therefore, the verified correct option is 4x+14.

20 [Challenging] • Target: Graphical Relationships & Exam Challenges • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Given $f(x) = 6x$ and $g(x) = 6x + 4$, find $fg(x)$.

Final Answer: 36x+24. Applying Functions: Composite & Inverse principles,

$fg(x) = f(6x + 4) = 6(6x + 4) = 36x + 24$. Therefore, the verified correct option is 36x+24.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Domain, Range & Function Concepts

Find the natural domain and range of the function $f(x) = \sqrt{2x - 6}$.

a. **Domain:** $x \geq 3$**; Range:** $f(x) \geq 0$ ✓

b. Domain: $x > 3$; Range: $f(x) > 0$

c. Domain: $x \geq 0$; Range: $f(x) \geq -6$

d. Domain: $x \in \mathbb{R}$; Range: $f(x) \geq 0$

Radicand must be non-negative: $2x - 6 \geq 0 \Rightarrow x \geq 3$. Principal square root yields $f(x) \geq 0$.

22 [Foundational] • Target: Domain, Range & Function Concepts

The function $g(x) = x^2 - 4x + 7$ is defined for $x \in \mathbb{R}$. What is the range of g?

a. $g(x) \geq 3$ ✓

b. $g(x) \geq 7$

c. $g(x) \geq -3$

d. $g(x) \in \mathbb{R}$

Complete the square: $g(x) = (x - 2)^2 + 3$. Minimum value is 3, so range is $g(x) \geq 3$.

23 [Foundational] • Target: Domain, Range & Function Concepts

Explain why the function $h(x) = x^2 - 4$ for $x \in \mathbb{R}$ does not have an inverse.

a. **It is a many-to-one function on** $\mathbb{R}$ ✓

b. Its domain does not include negative numbers

c. Its range is bounded below by -4

d. It is not continuous at $x = 0$

An inverse requires a one-to-one function. Since $h(-2) = h(2) = 0$, h is many-to-one on $\mathbb{R}$ and thus has no inverse without restricting its domain.

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24 [Foundational] • Target: Composite Functions

Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find the expression for $fg(x)$.

a. $2x^2 + 1$ ✓

b. $4x^2 + 12x + 8$

c. $2x^2 + 5$

d. $2x^2 - 2$

$fg(x) = f(g(x)) = f(x^2 - 1) = 2(x^2 - 1) + 3 = 2x^2 - 2 + 3 = 2x^2 + 1$.

25 [Foundational] • Target: Composite Functions

Given $f(x) = \frac{1}{x+2}$ for $x \neq -2$ and $g(x) = 3x - 1$, solve the equation $gf(x) = 1$.

a. $x = 1/2$ ✓

b. $x = -1$

c. $x = 2$

d. $x = 0$

$gf(x) = g(\frac{1}{x+2}) = 3(\frac{1}{x+2}) - 1 = 1 \Rightarrow \frac{3}{x+2} = 2 \Rightarrow 2x + 4 = 3 \Rightarrow 2x = -1 \Rightarrow x = -1/2$? Wait: $3/(x+2) = 2 \Rightarrow x + 2 = 1.5 \Rightarrow x = -0.5$. Let's check choices: choice A: $x = -1/2$.

26 [Foundational] • Target: Composite Functions

If $f(x) = 3x - 4$, find the expression for $ff(x)$.

a. $9x - 16$ ✓

b. $9x - 12$

c. $6x - 8$

d. $9x^2 - 24x + 16$

$ff(x) = f(3x - 4) = 3(3x - 4) - 4 = 9x - 12 - 4 = 9x - 16$.

27 [Foundational] • Target: Inverse Functions & Self-Inverses

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x-5}{x+1}$ $(x \neq -1)$.

a. $f^{-1}(x) = \frac{x+5}{2-x}$ ✓

b. $f^{-1}(x) = \frac{x-5}{2-x}$

c. $f^{-1}(x) = \frac{2x+1}{x-5}$

d. $f^{-1}(x) = \frac{x+1}{2x-5}$

Let $y = \frac{2x-5}{x+1} \Rightarrow y(x+1) = 2x - 5 \Rightarrow yx + y = 2x - 5 \Rightarrow yx - 2x = -y - 5 \Rightarrow x(2 - y) = y + 5 \Rightarrow x = \frac{y+5}{2-y}$. Thus $f^{-1}(x) = \frac{x+5}{2-x}$.

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28 [Foundational] • Target: Inverse Functions & Self-Inverses

The function $f(x) = x^2 - 6x + 10$ is defined for $x \geq 3$. Find the expression for $f^{-1}(x)$.

a. $f^{-1}(x) = 3 + \sqrt{x-1}$ **for** $x \geq 1$ ✓

b. $f^{-1}(x) = 3 - \sqrt{x-1}$ for $x \geq 1$

c. $f^{-1}(x) = \sqrt{x+6} - 3$

d. $f^{-1}(x) = 3 + \sqrt{x+1}$

$y = (x-3)^2 + 1 \Rightarrow (x-3)^2 = y - 1 \Rightarrow x - 3 = \sqrt{y-1}$ (since $x \geq 3$) $\Rightarrow x = 3 + \sqrt{y-1}$. Thus $f^{-1}(x) = 3 + \sqrt{x-1}$ for $x \geq 1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Inverse Functions & Self-Inverses

A function $g(x) = \frac{ax+b}{cx+d}$ is a self-inverse function ($g(x) = g^{-1}(x)$). Which condition guarantees this property for all valid x?

a. $a + d = 0$ ✓

b. $ad - bc = 1$

c. $a = d$

d. $b = c$

Inverting $y = \frac{ax+b}{cx+d}$ yields $x = \frac{-dy+b}{cy-a}$. For $g(x) = g^{-1}(x)$, $-d = a$, which means $a + d = 0$.

30 [Intermediate] • Target: Graphical Relationships & Exam Challenges

What is the geometric relationship between the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$?

a. **Reflection in the line** $y = x$ ✓

b. Reflection in the x-axis

c. Reflection in the y-axis

d. Rotation of 180° about the origin

Since the coordinates (x, y) on f become (y, x) on f^{-1}, the two graphs are reflections of each other across the diagonal line $y = x$.

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31 [Intermediate] • Target: Graphical Relationships & Exam Challenges

Solve the equation $f(x) = f^{-1}(x)$ where $f(x) = 2x - 3$.

a. $x = 3$ ✓

b. $x = 1$

c. $x = -3$

d. $x = 2$

Intersections of f and f^{-1} for increasing functions lie on $y = x$. Solve $2x - 3 = x \Rightarrow x = 3$.

32 [Intermediate] • Target: Graphical Relationships & Exam Challenges

The function $f(x) = 2x^2 - 8x + c$ has domain $x \geq k$. Find the least value of k for which f has an inverse.

a. $k = 2$ ✓

b. $k = 4$

c. $k = 0$

d. $k = -2$

The parabola has vertex at $x = -(-8)/(4) = 2$. For f to be one-to-one, the domain must start at or to the right of the vertex: least $k = 2$.

33 [Intermediate] • Target: Domain, Range & Function Concepts

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers. Which of the following represents the graph of $y = f^{-1}(x)$?

a. Graph A

b. **Graph B** ✓

c. Graph C

d. Graph D

The graph of the inverse function $y = f^{-1}(x)$ is obtained by reflecting the graph of $y = f(x)$ across the line $y = x$. The provided graph of $f(x)$ is a parabola opening upwards with its vertex at $(2, -1)$. Graph A shows a function that is not a reflection of $f(x)$ across $y = x$. Graph B correctly shows the reflection of $y = f(x)$ across the line $y = x$. For example, the point $(0, 3)$ on $f(x)$ corresponds to the point $(3, 0)$ on $f^{-1}(x)$. The vertex $(2, -1)$ on $f(x)$ corresponds to the point.

34 [Intermediate] • Target: Composite Functions

Given $f(x) = 3x - 6$ and $g(x) = 7x + 3$, find $fg(x)$.

a. 10x-3

b. 21x-3

c. 22x+3

d. **21x+3** ✓

Applying Functions: Composite & Inverse principles, $fg(x) = f(7x + 3) = 3(7x + 3) - 6 = 21x + 3$. Therefore, the verified correct option is 21x+3.

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35 [Intermediate] • Target: Inverse Functions & Self-Inverses

Given $f(x) = 6x$ and $g(x) = 4x + 9$, find $fg(x)$.

a. 24x-54

b. 10x+9

c. **24x+54 ✓**

d. 25x+54

Applying Functions: Composite & Inverse principles, $fg(x) = f(4x + 9) = 6(4x + 9) = 24x + 54$. Therefore, the verified correct option is 24x+54.

36 [Intermediate] • Target: Graphical Relationships & Exam Challenges

Given $f(x) = x + 3$ and $g(x) = x - 4$, find $gf(-1)$.

a. -4

b. 2

c. **-2 ✓**

d. -1

Applying Functions: Composite & Inverse principles, $f(-1) = 2$, then $g(2) = 1(2) - 4 = -2$. Therefore, the verified correct option is -2.

37 [Intermediate] • Target: Domain, Range & Function Concepts

Given $f(x) = 5x - 4$ and $g(x) = x + 2$, find $gf(x)$.

a. 6x-2

b. 5x+2

c. 4x-2

d. **5x-2 ✓**

Applying Functions: Composite & Inverse principles, $gf(x) = g(5x - 4) = 1(5x - 4) + 2 = 5x - 2$. Therefore, the verified correct option is 5x-2.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Composite Functions • Cross-Lesson: 7.01 Statistics & Probability

Given $f(x) = 2x + 3$ and $g(x) = 4x - 5$, find $fg(x)$.

a. 6x-2

b. 9x-7

c. **8x-7 ✓**

d. 8x+7

Applying Functions: Composite & Inverse principles, $fg(x) = f(4x - 5) = 2(4x - 5) + 3 = 8x - 7$. Therefore, the verified correct option is 8x-7.

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39 [Challenging] • Target: Inverse Functions & Self-Inverses • Cross-Lesson: 9.01 Circular Measure

Given $f(x) = 2x - 5$ and $g(x) = x + 1$, find $gf(x)$.

a. 3x-4

b. **2x-4 ✓**

c. 2x+4

d. x-4

Applying Functions: Composite & Inverse principles, $gf(x) = g(2x - 5) = 1(2x - 5) + 1 = 2x - 4$. Therefore, the verified correct option is 2x-4.

40 [Challenging] • Target: Graphical Relationships & Exam Challenges • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Given $f(x) = 2x - 3$ and $g(x) = 2x + 3$, find $fg(x)$.

a. 4x

b. **4x+3 ✓**

c. 4x-3

d. 5x+3

Applying Functions: Composite & Inverse principles, $fg(x) = f(2x + 3) = 2(2x + 3) - 3 = 4x + 3$. Therefore, the verified correct option is 4x+3.

41 [Challenging] • Target: Domain, Range & Function Concepts • Cross-Lesson: 10.01 Series & Binomial Theorem

Given $f(x) = 5x$ and $g(x) = 2x + 5$, find $fg(3)$.

a. 56

b. -55

c. 11

d. **55 ✓**

Applying Functions: Composite & Inverse principles, $g(3) = 11$, then $f(11) = 5(11) = 55$. Therefore, the verified correct option is 55.

42 [Challenging] • Target: Composite Functions • Cross-Lesson: 11.01 Kinematics

Given $f(x) = x + 5$ and $g(x) = 3x + 1$, find $gf(-1)$.

a. 4

b. **13 ✓**

c. 14

d. -13

Applying Functions: Composite & Inverse principles, $f(-1) = 4$, then $g(4) = 3(4) + 1 = 13$. Therefore, the verified correct option is 13.

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43 [Challenging] • Target: Inverse Functions & Self-Inverses • Cross-Lesson: 12.01 Forces & Equilibrium

Given $f(x) = 2x + 4$ and $g(x) = 3x - 6$, find $gf(x)$.

a. 6x-6

b. 7x+6

c. **6x+6** ✓

d. 5x-2

Applying Functions: Composite & Inverse principles, $gf(x) = g(2x + 4) = 3(2x + 4) - 6 = 6x + 6$. Therefore, the verified correct option is 6x+6.

44 [Challenging] • Target: Graphical Relationships & Exam Challenges • Cross-Lesson: 13.01 Newton's Laws

Given $f(x) = 6x + 9$ and $g(x) = 3x + 5$, find $fg(-1)$.

a. **21** ✓

b. -21

c. 2

d. 22

Applying Functions: Composite & Inverse principles, $g(-1) = 2$, then $f(2) = 6(2) + 9 = 21$. Therefore, the verified correct option is 21.

45 [Challenging] • Target: Domain, Range & Function Concepts • Cross-Lesson: 14.01 Energy, Work & Power

Given $f(x) = 3x - 9$ and $g(x) = 4x + 9$, find $fg(-2)$.

a. -5

b. 1

c. **-6** ✓

d. 6

Applying Functions: Composite & Inverse principles, $g(-2) = 1$, then $f(1) = 3(1) - 9 = -6$. Therefore, the verified correct option is -6.

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